MAPÚA MALAYAN DIGITAL COLLEGE

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Requirements Analysis Document of MotorPH Payroll System

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1. Introduction

MotorPH has relied on fragmented manual processes, such as paper timecards, spreadsheet-based payroll calculations, and branch-specific data reconciliation, to drive its growing workforce across multiple locations, which led to delayed payslips, calculation errors, and general bottlenecks that draw out the series of processes caused through repetition due to minor revisions in the payroll. The MotorPH Payroll System is designed to simplify and automate the once convoluted end-to-end payroll process for MotorPH, encompassing attendance tracking, salary computation, leave management, and compliance to government contributions. Driven by the shortcomings identified in the Business Requirement Specification (BRS) and the Current State Assessment of existing manual workflows, the system addresses inefficiencies, reduces processing times, and enhances data accuracy.

1.1. Purpose

This document provides a detailed description of how the MotorPH Payroll System and its components function and interact, serving as the basis and blueprint for the development, implementation, and maintenance of the said system.

1.2. Scope

The MotorPH Payroll System is designed to support the core administrative processes essential to compensating employees accurately, securely, and on time, primarily covering the digital management of employee records, attendance tracking through both remote and on-site channels, and the smoother handling of payroll computations, deductions, and reporting, whilst also accommodating for basic leave coordination and internal communication surrounding payroll-related matters, all within a secure, role-aware environment that respects data sensitivity and organisational hierarchy.

Nevertheless, despite the system being built to scale with the company's growing needs, it remains focused on the aforementioned detailed purposes, intentionally excluding non-payroll functionalities such as sales tracking, performance evaluation, or customer management. Likewise, it will not account for older historical records beyond a defined threshold nor will the project provision physical infrastructure for its integration. Its intent is not to replace

broader management tools but rather to deliver a focused, dependable solution for the organisation's payroll ecosystem.

1.3. Definitions Acronyms and Abbreviations

Term	Definition
BRS	**Business Requirement Specification** – a document outlining the business needs and expectations.
RAD	**Requirements Analysis Document** – outlines system objectives, scope, and specifications.
HR	**Human Resources** – the department responsible for managing employee relations and data.
PR Staff	**Payroll Staff** – personnel tasked with salary computation, reporting, and distribution.
2FA	**Two-Factor Authentication** – a login method requiring two separate forms of identification.
QR Code	**Quick Response Code** – a machine-readable code used for on-site attendance validation.
PII	**Personally Identifiable Information** – sensitive personal data such as TIN, SSS, PhilHealth numbers.
SSS	**Social Security System** – government-mandated employee benefit and contribution programme.
PhilHealth	**Philippine Health Insurance Corporation** – national health insurance system.
Pag-IBIG	**Home Development Mutual Fund (*Pagtutulungan sa Kinabukasan: Ikaw, Bangko, Industriya at Gobyerno)*** – government-run savings programme for housing and loans.
RBAC	**Role-Based Access Control** – security system that limits user access based on their organisational role.
PTO	**Paid Time Off** – compensated leave granted to an employee.

Ticketing	Earliest records of such a system being used is usually attributed to IBM. A ticket signified “your item (concern) is in the queue.” Internal request system used to track submissions and resolutions across HR, payroll, and admin.
On/Offboarding	The structured process of accepting/exiting an employee from the organisation.

1.4. References

Reference No.	Reference Document
1	Business Requirement Specification Document
2	Current State Assessment Report
3	List of Functional and Nonfunctional Requirements
4	Venn Diagram of System User Requirements

2. Stakeholder Identification

- **Stakeholder identification:**
  - **System Administrator:** Oversees system configuration, security policies, and maintenance.
  - **HR Personnel:** Handles employee profiles and onboarding/offboarding.
  - **Payroll Staff:** Supervise payroll calculation and distribution.
  - **Immediate Supervisors:** Team management in timesheets, schedules, attendance, leave, and overtime approval**.**
  - **Employees:** View personal attendance, estimated payslips, and submit tickets for notice or assistance.
- **Stakeholder needs:**
  - **System Administrator**: High system uptime, audit logs, overall and granular access control.
  - **HR Personnel**: Quick search and update of employee data, onboarding/offboarding workflow.
  - **Payroll Staff**: Automated payroll, specific and mass handling of adjustments for payslips, dispute tracking, monthly report generation.

- **Supervisors/Leads**: Team roster visibility, schedule adjustments, overtime and leave approvals, timesheet updates.
- **Employees**: Reliable clock-in/out (on-site and remote), personal attendance tracking, salary preview, ticket status updates, team roster and leave calendar.

3. Current System Overview

As mentioned previously, MotorPH's current payroll operations rely heavily on manual methods, resulting in a fragmented and error-prone system:
Attendance is logged through physical timecards and punch clocks, which limits scalability and excludes support for remote or offsite work, with such reliance on paper-based logging causes and increases the risk of lost or damaged records, creates unnecessary workload for supervisors, and lacks an audit trail that management can reference during disputes or verifications.

Payroll data is managed across branches through spreadsheet-based timesheets that must be manually routed. These processes operate too sequentially, with each approval or correction introduces backsteps and delays, especially when disputes arise. Timesheet and incentive inconsistencies require a back and forth between employee and leads and employee information updates require manual intervention by HR and until these updates are completed, affected payroll entries are frozen, potentially delaying the issuance of payslips not only for one employee but for an entire team.

Another major issue is the absence of real-time integration across departments. The current lack of a centralised access hub for tasks that rely on other branches leads to miscommunication, redundancy, and increased waiting times for payroll approval.

Additionally, security practices lack sophistication. Sensitive personally identifiable information (PII) is potentially open just to anyone in HR, exposing MotorPH to prospective privacy violations. There is also no dispute tracking system to monitor the frequency or resolution of employee concerns, resulting in repeated cycles of the same issues.

These challenges alone make it clear that the current system is not sustainable as MotorPH expands to multiple physical branches. A centralised, management and

automated payroll system is not just a convenience but a necessity to ensure accuracy, operational efficiency, and regulatory compliance.

4. Proposed System Overview

4.1. Overview

The MotorPH Payroll System is a modern, integrated platform developed to streamline compensation processes, support workforce mobility, and promote operational clarity across the organisation. By combining secure logins with branch-specific access and two-factor authentication, the system ensures that each user is connected to the right page at the right time.

Employees can record their time through mobile or on-site methods, view detailed attendance logs, and access projections of their upcoming earnings. A built-in request system facilitates formal communication for leaves, payroll queries, and support, with each submission tracked transparently.

Supervisors are equipped with tools to manage team availability and respond to employee requests, while HR staff can oversee personnel data and transitions with ease. And lastly, the primary aspect in need of a solution: the payroll module automates salary computation, integrates deductions and incentives, and ensures timely distribution of electronic payslips.

4.2. Functional Requirements

Req. ID No.	Name	Requirement Description
1	Employee Login	● Employees can login with their valid credentials ● Needs 2FA alongside with their credentials available through an app ● Redirect to the branch page respective of their Employee ID
2	Clock In	● Employees can time in on their app and will be registered as clocked in but offsite ● A QR code will pop up on the side which will update clock in status once scanned onsite ● Clocking out is also available in the app

		○ Clocking out will have a pop up to check if you're really clocking out ○ Clocking out before your hours are up will bring up another pop up
3	Employee Attendance	● Monitor the following personal attendance metrics from a certain date range (with default being earliest to latest): ○ Time in and out log ○ Days absent, present, and tardy ○ Hours overtime and undertime ○ Leave credits and leave taken
4	Salary Tracker	● Employees can take a peek at what likely would be, for the current payroll cycle, their position, hourly wage, accumulated regular, overtime, pto, and holiday overtime hours, total incentives, gross pay, income tax, other deductibles, and net pay; this is so disputes can be submitted and resolved earlier
5	Tickets	● Forms for overtime, leave, and timesheet update requests (sent to leads) ● Forms for payroll dispute (sent to payroll staff) ● Forms for general inquiries and qualms (sent to HR) ● Forms for Tech Support (sent to admin) ● Tickets will have a log to detail each ticket sent: ○ When it was sent ○ How far are you in the queue ○ Ticket resolution status ○ Reference ID for the ticket ● Tickets can be cancelled by the employee and it will be removed from the queue it was assigned to
6	Leave Tracker	● Employees can see the team roster where they can see the contact info of their colleagues and see their availability ● Employees can see the leave calendar to see when team members are having their leave to

		allow for better planning
7	Team Management	● Leads can see team roster where they can see the contact info of their members, and availability and attendance ● In the roster, the leads can adjust their team members' schedules ● Leads can see and address pending tickets, what type, when it was submitted, and who submitted it
8	HR Handling	● HR can see the overall employee statistics ● HR can see details of and manage those within the onboarding and offboarding queues ● HR can see and manage pending tickets ● HR can search and modify existing or add new employee records
9	Payroll Supervision	● Calculations will be done through the following: ○ Wage will be calculated through the retrieval and tally of employee rate, overtime, PTO, holiday, and rest day pay ○ Incentives will be calculated through the retrieval of rice subsidy, phone and clothing allowance, and other incentives ○ Deductibles will be calculated through the retrieval of government statutory contributions, withholding tax, and other deductibles ● Changes on rates for withholding tax and government contributions are accomplished here ● PR Staff can add incentives and deductibles to grouped members (global, team, branch, etc.) or specific members
10	Payroll Distribution	● Payslips are emailed to employee recipients for review at the end of pay cycle ● Payroll report generation for PR staff to

		review when payroll disputes are resolved

4.3. Non-Functional Requirements

Beyond simply delivering its intended features, the system must operate with a level of responsiveness, stability, and security that users across all organisational roles can consistently rely upon. Developed to address the inefficiencies of a divided, manual setup, the system is expected to perform routine operations promptly and without friction, regardless of growing data volumes or concurrent activity. It should uphold the limited accessibility of sensitive information through structured access controls, whilst remaining resiliently operative in the face of technical interruptions. Altogether, the standard it must reach is not only functional adequacy but one of confidence: that every user, at any given time, can trust the system to respond quickly, handle information securely, and support monitoring and approvals without compromise.

4.3.1. Performance

- System must respond quickly to employee and admin actions.
- Clock-in, ticket submissions, and salary viewing should have minimal latency.
- Payroll computations and generation of payslip reports must not exceed more than a few seconds under standard load.

4.3.2. Scalability

- System can handle up to at least 100 concurrent employee records (example record gave 34) and expand proportionally without performance loss.
- Should accommodate data growth through archival of older logs and separation of hot and cold data to prevent query overload.

4.3.3. Security

- 2FA in place for login
- Government Personally Identifiable Information (PIIs) are not accessible unless approved by two of either PR or HR staff
- RBAC enforcement for access and general visibility of specific delegated tasks and functionalities

4.3.4. Reliability

- Payroll calculations, leave logs, and clock-in timestamps should never be lost due to unexpected shutdowns, ensured by backup and recovery protocols.

5. Approval and Sign-Off

We, the undersigned, have reviewed the contents of this Requirements Analysis Document and hereby provide our approval and sign-off, indicating our agreement with the stated requirements for the MotorPH Payroll System.

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